



## In This Issue

This issue offers three papers selected for expansion from the International Symposium of Empirical Software Engineering (ISESE) 2004 as well as two regular submitted papers. The program chairs for ISESE04 were Natalia Juristo and Forrest Shull, two of our editorial board members.

2004 marked only the third anniversary of the Symposium and the first time it was located in the United States. Despite being such a young conference, ISESE has quickly become an established part of the international research community in empirical software engineering. The strength of the conference programs that we have seen to date is reflected in the large number of returning authors who submitted papers in 2004, as well as the continued growth in the number of submitters, indicating that new researchers are interested in becoming involved. Like all ISESE programs, 2004 contained a mix of papers describing empirical studies of various software development practices as well as papers on empirical methodologies themselves.

The papers that were selected for this special issue were picked by first asking the program committee for recommendations from the papers that they reviewed. From this set, we selected the three with the overall highest recommendations from all reviewers. The authors were then approached and asked to significantly expand their conference papers into versions suitable for journal publication.

In the first such paper, Do, Elbaum, and Rothermel present a framework of reusable artifacts and data, designed to decrease the cost and difficulty of experimentation with software test techniques. The challenges that are faced by experimenters in this area of software engineering are described and used to motivate the infrastructure that has been developed. Finally, empirical results are presented that demonstrate that the infrastructure can be used by researchers and describe the benefits obtained.

The second ISESE special issue paper also happens to focus on software testing but from the point of view of the development community rather than researchers. Vegas and Basili present a characterization schema that has been created to allow software developers to pick optimal test techniques for a given problem. The paper presents a survey of the literature and details the development of the schema, and its population from publications and expert knowledge. An experimental evaluation is described that investigated the effectiveness and usability of schema use.

In the third ISESE paper, Verelst investigates basic software phenomenology by empirically studying the influence of the abstraction level of a conceptual model on its evolvability. The impact of abstraction level was investigated with respect to the time needed to apply a change, the correctness of the change, and the structure degradation incurred. These results have important implications for work related to models of software systems in areas such as software architectures, reuse, design patterns, and application frameworks.

In the first of our two regular submissions, Judith Segal identifies two problems with the development of a set of software library components by software engineers for scientists: the limitations of communicating through requirements and the inability to evolve the requirements based upon experience. She uses the case study to argue for a methodology that integrates sequential and agile software development methods. The paper provides some first hand insight into the problems of communication using traditional development methods.

In the second regular paper, Takagi, Mizuno, and Kikuno, use a questionnaire to characterize the level of confusion in a project, linking the measure of confusion to the inability of the project to detect potential risks. They developed a methodology based upon an analysis of the questionnaire, to recognize which projects were in a state of confusion and thereby did not identify risks appropriately. They applied the method to over three dozen projects and were able to differentiate risky projects from non-risky projects almost 90% of the time. The paper offers an approach for identify projects at risk.

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